Beyond Content:

Temporal Structure and Distinct Coupling Regimes in Two Human-LLM Archives

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April 2026

Abstract

Prior work in this series established that long-form human-LLM interaction can exhibit organized temporal structure (Baldovino, 2026a) and that stateless, surface-matched synthetic baselines do not fully reproduce it (Baldovino, 2026b). The present paper asks whether that organization generalizes across archives by comparing two matched human-LLM archives. Using three paired threads exceeding 600 turns, processed in the same export format and truncated within each pair to a matched length, the analysis compares structural metrics only: turn lengths, rolling variability, amplification, consecutive-turn correlation, and separated user and assistant rolling means. The three pairs support the same higher-order conclusion even where the separating metric profile differs across pairs. What generalizes is not a single shared morphology but the presence of temporally organized long-form regimes whose realization differs across human-coupled archives.

1. Introduction

The present paper extends a two-step prior argument. *Beyond Content: Temporal Dynamics of Long-Form Human-LLM Interaction* (Baldovino, 2026a) showed that extended human-LLM archives can exhibit large-scale temporal structure. *Beyond Content: Surface Matching, Temporal Divergence, and the Limits of Synthetic Baselines* (Baldovino, 2026b) showed that a stateless, surface-matched synthetic null did not fully recover that structure. Together, these establish organized structure in one archive that is not reducible to surface-matched autoregressive generation. What remains open is whether the organization observed in Paper 1 is a property of that specific archive or a feature of temporally organized long-form interaction more broadly. The comparison therefore moves from archive-plus-null to archive-versus-archive.

The question is not whether the two archives differ in obvious surface ways, but whether temporally organized long-form interaction is expressed through the same coupling morphology across archives. The argument is restrained: what may generalize is not one fixed morphology, but the presence of organized long-form regimes whose realization differs across human-coupled archives.

2. Research Question and Predictions

Do two matched long-form human-LLM archives exhibit a shared class of temporal organization while differing in the coupling structure through which that organization is expressed?

Prediction 1 (null). Under structural metrics alone, two matched long-form human-LLM archives are not distinguishable beyond broad surface variation. Any observed differences reduce to local fluctuation and do not support distinct coupling morphology once topic, tone, and lexical content are excluded.

Prediction 2 (working). Under structural metrics alone, the two archives remain distinguishable on one or more coupling-level measures, including user length distribution, rolling variability, amplification, and consecutive user-turn correlation. Across the three matched pairs, the same general pattern holds even where the separating metric profile differs: both archives exhibit temporally organized long-form structure, while the coupling regimes through which it is expressed are archive-specific. This does not imply model learning, agency, or semantic interpretation.

3. Methods

Three paired long-form threads exceeding 600 turns were selected across Human A and Human B. All threads were drawn from the same model family and processed using the same HTML export format.

Matched dimensions across pairs include model family (GPT-5, with minor version drift common to both archives via ChatGPT's automatic model updates during the capture period), interface (ChatGPT app), and user familiarity (both humans are experienced long-form users of the model). Thread length is matched within each pair by truncation to the shorter of the two. Conversation type and topic were not matched, for the reason described below.

The two archives differ substantially in conversation content. Human A's threads are predominantly research- and theory-oriented (analysis, methodology development, extended reasoning). Human B's threads span everyday discursive use (household questions, personal interests, open-ended discussion). This content asymmetry is deliberate. Within the Beyond Content framework, structural metrics should remain interpretable across archives with divergent semantic content if the non-semantic claim is robust. The comparison therefore offers a more demanding test of the non-semantic claim than a topic-matched comparison would: if structural metrics continue to separate the archives across divergent content categories, the framework's central commitment is preserved under higher content variance. The design trades this analytical benefit against the loss of control that a topic-matched comparison would provide.

The analysis is intentionally structural: no semantic content analysis, topic modeling, or tone coding is used. The figure set evaluates archive-level separation using user length distribution, assistant rolling standard deviation, normalized amplification ratio, consecutive user-turn

correlation, and separated user and assistant rolling means. The comparison is therefore confined to temporal organization and coupled structural behavior rather than meaning-level interpretation.

3.1 Paired-thread construction

Pair	Human A turns	Human B turns	Truncated length
1	714	752	714
2	878	768	768
3	2,577	1,158	1,158

All paired analyses are conducted on the truncated length for each pair. This preserves within-pair comparability and does not imply that longer-thread continuation is uninformative.

3.2 Metric-to-claim map

Metric	What it tests	Interpretive read
User length distribution	Whether simple structural summaries already separate the archives	Useful first-pass separation at the user-side envelope
Assistant rolling standard deviation	How volatility unfolds across normalized position	The archives differ in the breadth and placement of assistant-side variability
Amplification ratio	How user/assistant scaling evolves over normalized position	Tests archive-level differences in coupling expansion
Consecutive user-turn correlation	Local user-side continuity profile	Short-range user-side dependence profile
Separated user/assistant rolling means	Two-channel interaction profile across normalized position	Provides an archive-level overview of the two interaction profiles

4. Results

The results support a restrained conclusion. The three paired comparisons vary in which metrics separate the archives most strongly. This variation is itself consistent with the central claim: organized long-form regimes recur across archives, but their metric-level realization is pair-specific. The relevant question is whether both archives show temporally organized structure and whether structural differences between them remain visible across pairs.

4.1 User-side envelope comparison

The first comparison examines user-length distribution across the paired threads to assess whether the two archives differ at the level of the user-side structural envelope.

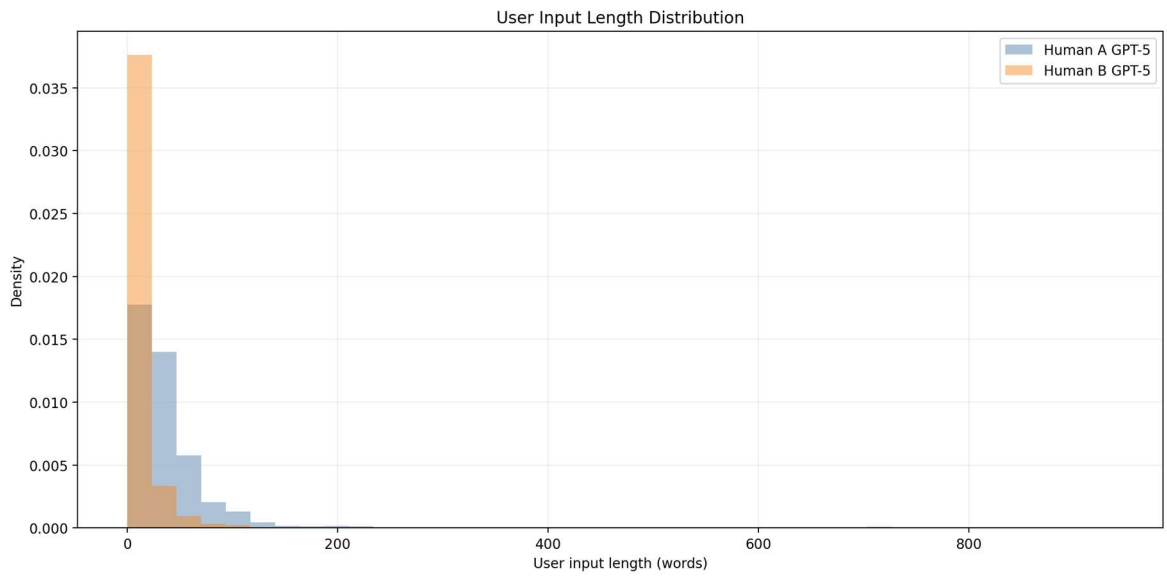

Figure 1. User-length distribution across the three paired threads. The figure provides an initial comparison of the user-side structural envelope in the two archives.

4.2 Rolling variability in assistant behavior

We next compare assistant-side rolling standard deviation across normalized thread position to evaluate differences in how variability is distributed over the course of the interaction.

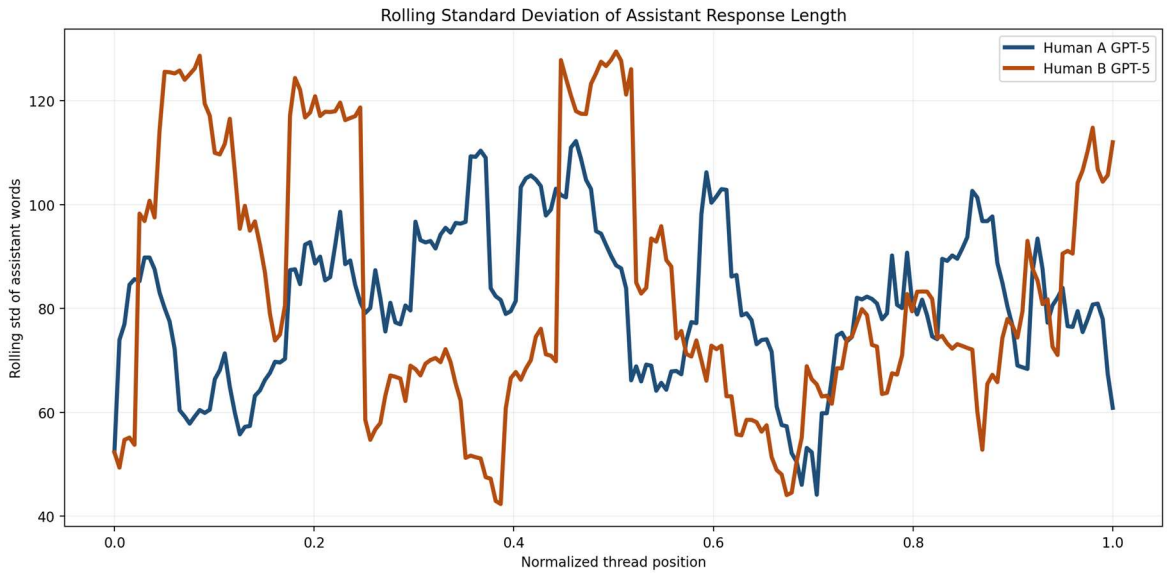

Figure 2. Assistant rolling standard deviation across normalized thread position. The figure shows distinct temporal profiles of assistant-side variability in the two archives.

4.3 Amplification as a coupling metric

Amplification ratio is examined here as a structural measure of how user-assistant scaling evolves across the interaction.

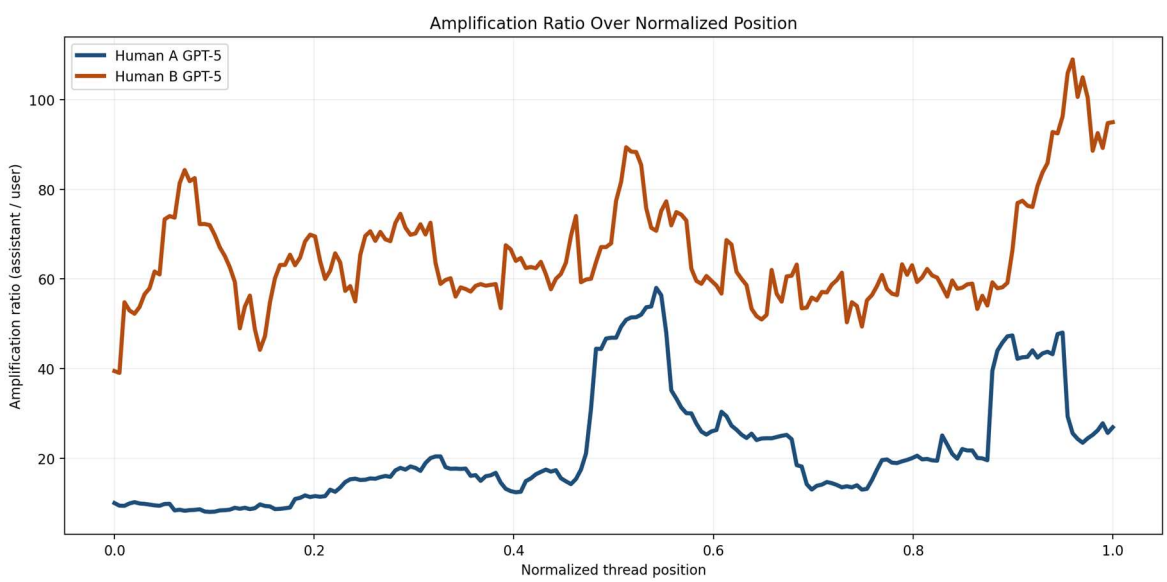

Figure 3. Normalized amplification ratio across paired threads. The two archives show distinct profiles of user-assistant scaling across normalized position.

4.4 Consecutive user-turn correlation

Consecutive user-turn correlation is used here to compare short-range user-side dependence across normalized thread position in the two archives.

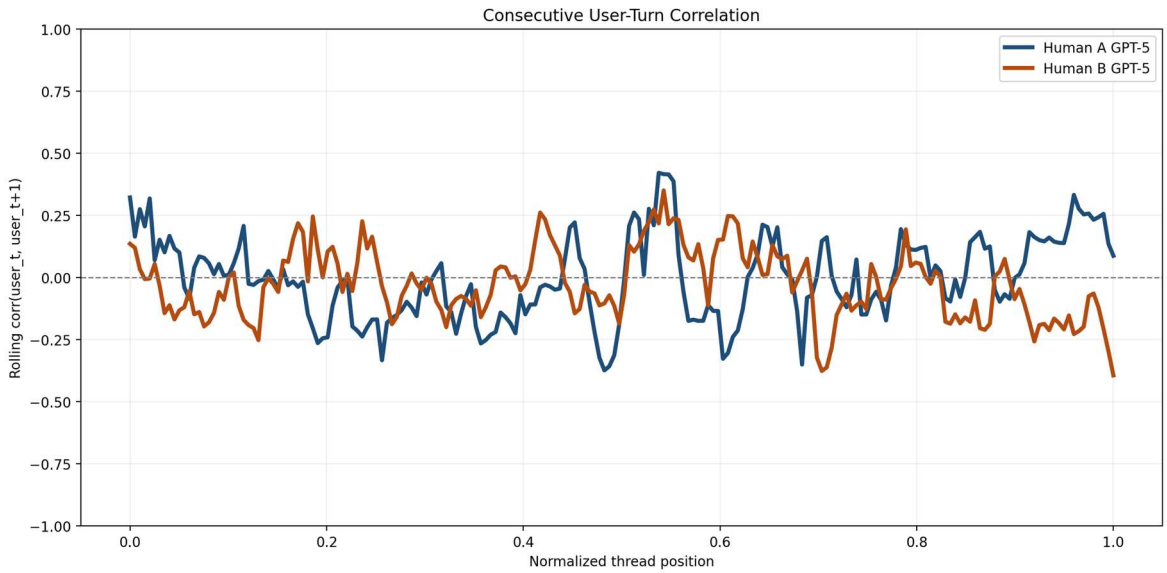

Figure 4. Consecutive user-turn correlation across normalized position. The two archives differ in their short-range user-side correlation profile.

4.5 Separated user and assistant rolling means

We close the figure set with separated user and assistant rolling means, shown together to provide an archive-level view of the two interaction profiles.

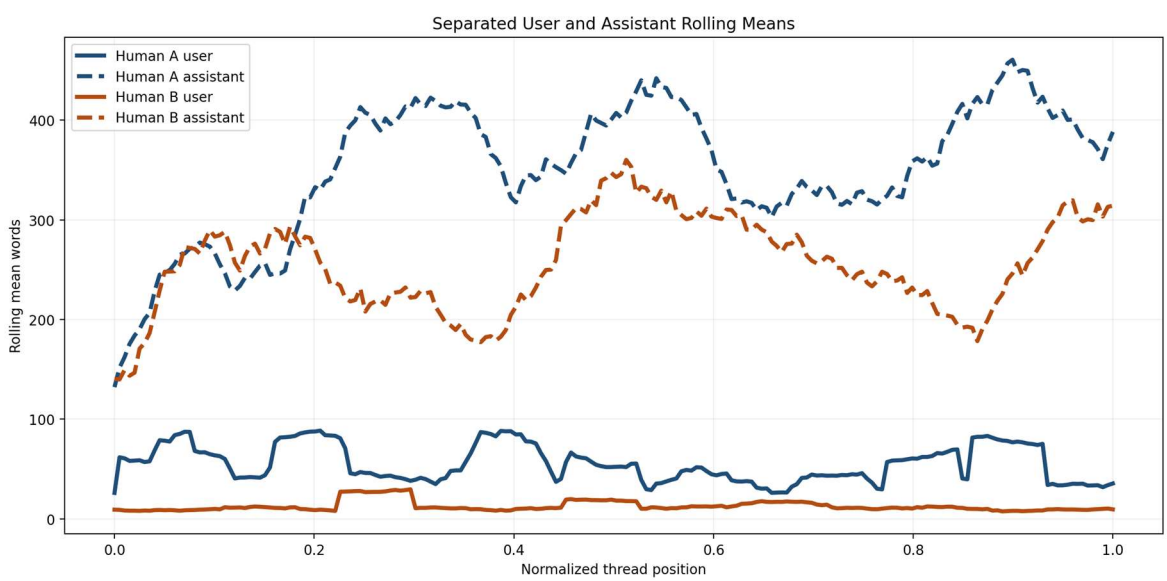

Figure 5. Separated user and assistant rolling means aggregated across the three paired threads. The figure provides an archive-level overview of the two interaction profiles.

5. Discussion

The results support a narrower claim than simple surface difference. Both archives exhibit temporally organized long-form structure, but the coupling profiles through which that structure is expressed differ across archives.

The implication is narrower than a claim of one fixed long-form morphology. A more defensible conclusion is that organized long-form regimes may generalize across archives while the exact realization of those regimes remains archive-specific.

Taken together, the results support a conservative cross-archive claim. Large-scale temporal organization need not reduce to one fixed structural profile, and archive-level differences remain visible even after stripping away topic, tone, lexical content, and meaning. What recurs across archives is the presence of organized long-form regimes; what varies is the coupling through which they are expressed.

6. Limitations

This comparison includes two archives and three paired threads only. The claims should therefore remain archive-level and should not be reframed as person-types, population claims, or universal user classes.

The present comparison uses archives to which the author has direct access. This is consistent with prior work in the Beyond Content series and is appropriate for exploratory methodology development at this stage of the program. The non-semantic design constrains the role of analyst judgment by deriving all comparisons from predefined structural metrics rather than interpretive coding. Independent replication with externally held archives is planned as a direct extension of this work.

Figure 5 (separated user and assistant rolling means) is aggregated across all three pairs rather than presented per-pair. Aggregation is appropriate for the archive-level overview it supports but obscures pair-level variation that the other figures preserve. The comparison is also fixed-boundary and same-model-family, which helps reduce obvious confounds but does not eliminate interface- or session-level differences that may vary within or across pairs.

Content variance is a design feature rather than a controlled variable. This is consistent with the non-semantic commitment of the Beyond Content framework, but it does leave open a specific alternative interpretation: that some portion of the observed coupling differences reflects systematic content-category effects (e.g., research discourse producing different temporal envelopes than discursive conversation) rather than person-level coupling style. Distinguishing these is beyond the scope of the present paper and is a natural extension for future work — specifically, within-person comparison across content categories, or cross-person comparison within a matched content category.

7. Conclusion

Across three matched long-form pairs, both archives exhibit temporally organized interaction, but the coupling profiles differ. The paired comparisons do not separate in identical ways, nor is that required for the central result to hold. What generalizes in this comparison is temporally organized long-form structure, not one fixed profile.

8. Author Contributions & AI Assistance Disclosure

This manuscript presents original theoretical work and conceptual modeling by Kristina Baldovino. The author is solely responsible for the research design, analytical synthesis, and interpretive conclusions. Portions of the manuscript text, including grammar refinement and formatting, were prepared with assistance from an OpenAI and an Anthropic language model under the author's direct guidance. The models served only as a technical writing aid. All intellectual content and editorial decisions remain the sole responsibility of the author.

9. References

Baldovino, K. (2026a). *Beyond Content: Temporal Dynamics of Long-Form Human-LLM Interaction* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18273459

Baldovino, K. (2026b). *Beyond Content: Surface Matching, Temporal Divergence, and the Limits of Synthetic Baselines* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19143515

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